

# Songeun Kim (Olivia)

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*Interests: Trustworthy generative AI, uncertainty quantification*

*I am particularly interested in studying generative models as stochastic decision-making systems, focusing on how randomness shapes model behavior, reliability, and controllability.*

## Education

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| 2022 – Present | <b>Seoul National University</b><br><i>Major: Statistics</i> <ul style="list-style-type: none"><li>GPA 3.94/4.30; 96.4%</li><li>Relevant Coursework:<br/><i>Probability and Measure Theory, Stochastic Differential Equations, Topics in Applied Mathematics: RL &amp; LLM(graduate), Nonparametric Function Estimation(graduate), NLP, Basics of Deep Learning, Bayesian Statistics</i></li></ul> | Seoul, Korea   |
| 2020 – 2021    | <b>Korea Advanced Institute of Science and Technology</b> <ul style="list-style-type: none"><li>Coursework in Dept. of Biological Sciences</li></ul>                                                                                                                                                                                                                                               | Daejeon, Korea |
| 2018 – 2019    | <b>Hansung Science High School</b> <ul style="list-style-type: none"><li>Early graduation due to excellence in Mathematics &amp; Science</li></ul>                                                                                                                                                                                                                                                 | Seoul, Korea   |

## Publications

[1] **Songeun Kim**, Seungyoo Lee, Edwin Fong, Hyungi Lee, Juho Lee, *From Drift to Coherence: Stabilizing Beliefs of LLMs*, ICML 2026, [arxiv](#)

## Research Experience

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |
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| Jun. 2026<br>– Present   | <b>Machine Learning Lab (MLLAB, SNU)</b><br><i>Undergraduate Research Intern</i> <ul style="list-style-type: none"><li>Advised by Prof. <a href="#">Hyunoh SONG</a></li></ul>                                                                                                                                                                                                                                                                                                                                                                          | Seoul, Korea |
| Jul. 2025<br>– Jan. 2026 | <b>Statistical Inference and Machine Learning Lab (KAIST AI)</b><br><i>Undergraduate Research Intern</i> <ul style="list-style-type: none"><li>Advised by Prof. <a href="#">Juho LEE</a></li><li>Investigated belief drift during LLM answer resampling and observed self-stabilizing dynamics over repeated sampling.</li><li>Reduced belief drift and improved predictive coherence on standard MCQA benchmarks(CSQA, AI2-ARC, TinyMMLU) with no loss in baseline accuracy via self-consistency fine-tuning loss and prompting strategies.</li></ul> | Seoul, Korea |
| Dec. 2024<br>– Jun. 2025 | <b>High-Dimensional Multiple Testing Lab (SNU)</b><br><i>Undergraduate Research Intern</i> <ul style="list-style-type: none"><li>Advised by Prof. <a href="#">Junyong PARK</a></li><li>Increased true-positive discoveries by 175% with an adaptive FDR-control algorithm for zero-inflated data (EM algorithm on Gamma mixture).</li><li>Built forward-selection covariate selection for high-dimensional data to surfact informative variables.</li></ul>                                                                                            | Seoul, Korea |

## Selected Projects

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| Fall 2025 | <b>Representational Bias Analysis in Toxicity Classification</b><br><i>Team leader</i> <ul style="list-style-type: none"><li>Introduced geometry-based metrics (toxicity-alignment score, subgroup pairwise distance measure) that expose latent identify bias not captured by aggregate accuracy or confusion-matrix based metrics.</li><li>On the Jigsaw benchmark, showed identity groups with higher FPR exhibit stronger alignment with toxicity directions and separate further BERT embedding space; tested contrastive and pairwise-distance regularization objectives for mitigating identity-term shortcut learning.</li></ul> |
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| Spring 2025 | <b>Lightweight Modular Fine-Tuning of Qwen-3B for ARC-AGI Tasks</b>                                                                                                                                                                              |  |
|             | <ul style="list-style-type: none"> <li>Fine-tuned Qwen-3B on ARC-AGI tasks using Unsloth, dividing tasks by shape transition rules and applying double adapters with lightweight data augmentation and Cartesian prompt reformatting.</li> </ul> |  |

## Work Experience

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| Feb. 2026   | <a href="#">LinqAlpha</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Seoul, Korea |
| – Jun. 2026 | <i>Forward Deployed Engineer</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |
|             | <ul style="list-style-type: none"> <li>Designed and optimized multi-agent workflows for financial analysis systems, focusing on agent role decomposition, tool orchestration, and controlled information flow across heterogeneous financial, market, and news data sources.</li> <li>Developed equity-research agent, including a temporally grounded news-search algorithm that constructs event timelines by aligning news publication timestamps with event occurrence dates.</li> </ul> |              |

## Teaching Experience

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|-------------|--------------------------------------------------------------------------------------------------------------------------------|--|
| Sep. 2024   | <b>Introduction to Modern Statistics Lecture (SNU)</b>                                                                         |  |
| – Dec. 2025 | <ul style="list-style-type: none"> <li>Mentored undergraduate students in modern statistics across three semesters.</li> </ul> |  |

## Awards and Scholarship

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| <ul style="list-style-type: none"> <li>Full-tuition scholarship (Dean’s list, College of Natural Science), Fall 2024 &amp; 2025</li> <li>R&amp;E excellence award in Physics &amp; Chemistry, all three semesters, 2018 – 2019</li> </ul> |  |
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## Technical Skills and Other Information

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| Languages       | <ul style="list-style-type: none"> <li>Native in Korean, proficient in English</li> </ul>                                                                                                                                           |
| Computer        | <ul style="list-style-type: none"> <li>Skilled in Python, R, PyTorch</li> </ul>                                                                                                                                                     |
| Extracurricular | <ul style="list-style-type: none"> <li>FIXERS (<i>Member</i>; Fixed Income &amp; Derivatives Research Society at SNU) in 2026</li> </ul>                                                                                            |
| Activities      | <ul style="list-style-type: none"> <li>LiberAct (<i>Stage Team leader</i>; SNU College of Liberal Arts Acting Club) in 2023</li> <li>SNU Habitat (<i>Member</i>; SNU Volunteer Club for Furniture Refurbishment) in 2022</li> </ul> |